Investigation of the continuum limit in the presence of small breaking of gauge invariance in lattice gauge theories

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I hereby declare that the work presented here was formulated by myself and that no sources or tools other than those cited were used.

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This thesis investigates whether and how the continuum limit of a lattice gauge theory is affected by small explicit violations of gauge invariance. The study combines theoretical scaling arguments with numerical measurements of Wilson loops, Creutz ratios, the static potential, the Sommer scale, and related dimensionless observables. The central goal is to determine whether gauge-breaking effects vanish, persist, or modify the extrapolation toward the continuum limit as the lattice spacing is reduced.

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Notation and Conventions

a Lattice spacing.

β Bare lattice coupling parameter.

ε_1 Bare parameter controlling the explicit breaking of gauge invariance.

$W(R,T)$ Wilson loop with spatial extent R and temporal extent T.

$\chi(R,T)$ Creutz ratio used to estimate the force and string tension.

r_0 Sommer scale defined through $r^2F(r) = 1.65$.

σ String tension.

1 Introduction

1.1 Context and Problem

Gauge invariance is one of the defining structural principles of lattice gauge theory. In Wilson's formulation it is preserved exactly at nonzero lattice spacing, which strongly constrains the allowed lattice artifacts and supports a controlled approach to the continuum limit. In practical and generalized settings, however, gauge invariance may be only approximate. Small explicit violations can arise from regulator artifacts, effective descriptions, algorithmic imperfections, or imperfect implementations of gauge constraints.

The central problem of this thesis is therefore not whether exact gauge invariance is useful, but how robust the continuum limit is when the lattice theory is perturbed by a controlled, small gauge-breaking deformation. This requires a comparison between the standard gauge-invariant theory and deformed ensembles along a sequence of lattice spacings.

1.2 Research Gap

The theoretical expectation depends on the scaling character of the gauge-breaking deformation. If the deformation acts like an irrelevant lattice artifact, its effect should vanish as the lattice spacing is reduced. If it contains a relevant component, it may change the continuum theory unless it is tuned away. For the linear single-link deformation used in this thesis, Chapter 3 shows that the weak-field expansion contains a relevant masslike component. The numerical study must therefore distinguish successful suppression along scaled trajectories from a genuine shift of continuum observables.

2 Scientific Background and Theoretical Framework

The aim of this chapter is to collect the theoretical ingredients needed for the analysis of the continuum limit in the presence of small explicit violations of gauge invariance. The discussion starts from continuum Yang–Mills theory and then introduces the lattice formulation, Wilson loops, the static potential, Creutz ratios, and the Sommer scale. These are the central concepts used later to define and compare physical observables at different lattice spacings.

2.1 Gauge Theories in the Continuum

2.1.1 Gauge Fields and Local Gauge Symmetry

[add citations] A non-Abelian gauge theory is built from fields that transform locally under a Lie group. In this thesis the relevant case is the group SU(2). A continuum gauge field can be written as

$$A_\mu(x) = A_\mu^a(x) T^a, \tag{2.1}$$

where T^a are the generators of the Lie algebra of SU(2). A common choice is $T^a = \sigma^a/2$, with σ^a the Pauli matrices. The gauge field defines the covariant derivative

$$D_\mu = \partial_\mu + iA_\mu, \tag{2.2}$$

and the associated field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]. \tag{2.3}$$

The commutator term distinguishes a non-Abelian gauge theory from an Abelian one. It implies that the gauge field itself carries charge under the gauge group and therefore self-interacts.

A local gauge transformation is specified by a group element

$$\Omega(x) \in \mathrm{SU}(2) \tag{2.4}$$

at every point in space-time. Under such a transformation, matter fields in the fundamental representation transform as

$$\psi(x) \mapsto \Omega(x)\psi(x), \tag{2.5}$$

while the gauge field transforms inhomogeneously,

$$A_\mu(x) \mapsto \Omega(x) A_\mu(x) \Omega^\dagger(x) + i\Big(\partial_\mu \Omega(x)\Big)\Omega^\dagger(x). \tag{2.6}$$

The inhomogeneous derivative term is necessary because the transformation is local. The field strength transforms covariantly,

$$F_{\mu\nu}(x) \mapsto \Omega(x) F_{\mu\nu}(x) \Omega^\dagger(x), \tag{2.7}$$

so traces of products of field strengths at the same point are gauge invariant.

In Euclidean space-time, the pure Yang–Mills action can be written as

$$S_{\mathrm{YM}} = \frac{1}{2g^2} \int \mathrm{d}^4x \, \mathrm{tr}\, F_{\mu\nu}(x) F_{\mu\nu}(x), \tag{2.8}$$

with summation over repeated Lorentz indices. This action is invariant under local gauge transformations. The coupling g controls the strength of the interaction. For an asymptotically free gauge theory, the effective coupling becomes small at short distances, whereas the long-distance theory is strongly coupled and requires non-perturbative methods.

2.1.2 Gauge-Invariant Observables

Physical observables in a gauge theory must be gauge invariant. Gauge-dependent quantities can be useful after gauge fixing, but their interpretation depends on the chosen gauge and is therefore not directly physical without additional care. For pure gauge theory, important gauge-invariant observables include traces of closed parallel transporters, local action-density operators, correlation functions of gauge-invariant operators, and quantities derived from Wilson loops.

A central continuum object is the path-ordered parallel transporter along a curve C,

$$U(C) = \mathcal{P} \exp\left(i \int_C \mathrm{d}x^\mu A_\mu(x) \right). \tag{2.9}$$

For a closed curve, the trace

$$W(C) = \frac{1}{N_c} \mathrm{Re}\, \mathrm{tr}\, U(C) \tag{2.10}$$

is gauge invariant. On the lattice this object becomes the Wilson loop and is one of the most direct probes of confinement and of the static quark–antiquark potential. [citation of source]

2.1.3 Physical Consequences of Gauge Invariance

Gauge invariance has direct consequences for the structure of the theory. It restricts the form of the action and selects the class of observables that can be interpreted without choosing a gauge. In the continuum, this is reflected in the Yang–Mills action, which is built from traces of the field-strength tensor. On the lattice, the same principle leads to actions and observables constructed from traces of closed products of link variables.

This restriction is important for the continuum limit. If gauge invariance is preserved exactly at finite lattice spacing, gauge-noninvariant terms are excluded by symmetry. The Wilson plaquette action implements this idea by using compact group-valued link variables and by forming the action from plaquette traces, i.e. from elementary closed loops [1, 2, 3]. As a result, the lattice theory has the correct local gauge symmetry before the continuum limit is taken.

For this thesis, the relevant question is what changes when this protection is relaxed. A gauge-breaking deformation can introduce additional terms that are not allowed in the gauge-invariant theory. If these terms are irrelevant in the renormalization-group sense, their effects vanish as the lattice spacing is sent to zero. If they contain relevant components, they can modify the continuum limit unless their coefficients are tuned or scaled away. The analysis in later chapters therefore compares gauge-invariant observables between the undeformed Wilson theory and ensembles with a controlled single-link deformation.

2.2 Lattice Gauge Theory

2.2.1 Discretization of Space-Time

In lattice gauge theory, Euclidean space-time is replaced by a hypercubic lattice

$$\Lambda_a = \left\{ x = an \,\middle|\, n \in \mathbb{Z}^4 \right\}, \tag{2.11}$$

where a is the lattice spacing. The lattice spacing acts as an ultraviolet cutoff. Momenta larger than order $1/a$ are not resolved. The continuum limit is obtained by sending

$$a \to 0 \tag{2.12}$$

while keeping physical quantities fixed.

Throughout this thesis the lattice is four-dimensional. Direction indices are written as $\mu, \nu = 1, 2, 3, 4$. Spatial directions are denoted by $i, j = 1, 2, 3$, while $\mu = 4$ is the Euclidean time direction. The integer temporal extent of a Wilson loop is denoted by T, and the temporal lattice extent is denoted by N_4.

In a numerical simulation the lattice has finite extent

$$L_\mu = N_\mu a, \tag{2.13}$$

where N_μ is the number of lattice points in direction μ. Therefore two limits are conceptually distinct: the continuum limit $a \to 0$ and the infinite-volume limit $L_\mu \to \infty$. In practice, simulations are performed at finite volume, and the lattice extents are chosen such that finite-volume effects are small compared with the statistical and systematic uncertainties relevant for the analysis.

In this thesis, the spatial volumes are chosen to keep the physical spatial extent approximately fixed where possible. The temporal lattice extent is kept fixed at $N_4 = 64$. Therefore the physical temporal extent $L_4 = N_4 a$ decreases as the lattice spacing is reduced. This is acceptable as long as the temporal extent remains large enough that finite-size effects in the time direction do not affect the observables used in the analysis. The main requirement is not that L_4 is held exactly fixed, but that it is large enough to allow a reliable extraction of Wilson-loop observables and, where needed, stable large-T behaviour.

2.2.2 Link Variables and Plaquettes

The fundamental degrees of freedom of Wilson lattice gauge theory are link variables

$$U_\mu(x) \in \mathrm{SU}(2), \tag{2.14}$$

which live on the oriented link from x to $x + a\hat{\mu}$. They are the lattice analogues of parallel transporters. For a smooth continuum gauge field, a link variable may be viewed as the parallel transporter along one lattice spacing,

$$U_\mu(x) \simeq \mathcal{P} \exp\left[i \int_x^{x+a\hat{\mu}} \mathrm{d}y^\mu \, A_\mu(y) \right]. \tag{2.15}$$

In the lattice path integral, however, the group-valued links $U_\mu(x)$, not the algebra-valued fields $A_\mu(x)$, are the fundamental variables.

To formulate local gauge invariance on the lattice, one assigns an independent group element to every lattice site. This makes it possible to test whether a given lattice expression depends only on gauge-invariant information or on the particular representative of a gauge orbit. This distinction is central for this thesis, because the Wilson plaquette action is gauge invariant, whereas the single-link deformation introduced later is not. A local lattice gauge transformation assigns a group element $\Omega(x)$ to each site and acts on the link variables as

$$U_\mu(x) \mapsto \Omega(x) U_\mu(x) \Omega^\dagger(x + a\hat{\mu}). \tag{2.16}$$

The transformation law shows why closed products of link variables are special. For an open path, the gauge transformations at the two endpoints remain. For a closed path, the endpoint is the same as the starting point, so the product transforms only by conjugation. Taking the trace then removes this remaining conjugation. The smallest nontrivial closed path on the lattice is the plaquette,

$$U_{\mu\nu}(x) = U_\mu(x) U_\nu(x + a\hat{\mu}) U_\mu^\dagger(x + a\hat{\nu}) U_\nu^\dagger(x). \tag{2.17}$$

Under a gauge transformation, the plaquette transforms covariantly at its base point,

$$U_{\mu\nu}(x) \mapsto \Omega(x) U_{\mu\nu}(x) \Omega^\dagger(x). \tag{2.18}$$

Therefore $\mathrm{Re}\,\mathrm{tr}\, U_{\mu\nu}(x)$ is gauge invariant [2]. This property is the reason why plaquettes can be used to build the gauge-invariant Wilson action.

For smooth fields, the plaquette has the continuum expansion

$$U_{\mu\nu}(x) = \exp\left[ia^2 F_{\mu\nu}(x) + \mathcal{O}(a^3)\right]. \tag{2.19}$$

Thus the plaquette measures the field strength through the holonomy around an elementary square. Inserting this expansion into the Wilson plaquette action shows that the leading nontrivial term is proportional to $a^4 \mathrm{tr}\, F_{\mu\nu}F_{\mu\nu}$. After summing over all plaquettes, the factor $a^4 \sum_x$ becomes the continuum space-time integral. This is why the Wilson plaquette action reproduces the Euclidean Yang–Mills action in the naive continuum limit [1, 2].

2.2.3 Wilson Gauge Action

The plaquette is the smallest closed loop that can be formed from link variables. Since the trace of a closed product of links is gauge invariant, the plaquette provides the simplest local building block for a gauge-invariant lattice action. The standard Wilson gauge action for $\mathrm{SU}(N_c)$ is

$$S_W[U] = \beta \sum_x \sum_{\mu<\nu} \left(1 - \frac{1}{N_c} \mathrm{Re}\,\mathrm{tr}\, U_{\mu\nu}(x)\right), \tag{2.20}$$

where each oriented plaquette plane is counted once. The bare lattice coupling is conventionally parametrized by $\beta = \frac{2N_c}{g_0^2}$. The constant term in Eq. (2.20) normalizes the action such that a trivial gauge field, $U_{\mu\nu}(x) = \mathbf{1}$, has zero action.

In this thesis the gauge group is SU(2). Therefore $N_c = 2$ and $\beta = \frac{4}{g_0^2}$. The Wilson action used as the undeformed reference action is thus

$$S_W[U] = \beta \sum_x \sum_{\mu<\nu} \left(1 - \frac{1}{2} \mathrm{Re}\,\mathrm{tr}\, U_{\mu\nu}(x)\right). \tag{2.21}$$

Because $U_{\mu\nu}(x)$ transforms by conjugation at its base point, the trace in Eq. (2.21) is invariant under local lattice gauge transformations. The Wilson action therefore preserves gauge invariance exactly at nonzero lattice spacing.

The connection to the continuum Yang–Mills action follows from the plaquette expansion. For a smooth continuum gauge field,

$$U_{\mu\nu}(x) = \exp\left[ia^2 F_{\mu\nu}(x) + \mathcal{O}(a^3)\right], \tag{2.22}$$

and hence

$$1 - \frac{1}{N_c} \mathrm{Re}\,\mathrm{tr}\, U_{\mu\nu}(x) = \frac{a^4}{2N_c} \mathrm{tr}\, F_{\mu\nu}(x) F_{\mu\nu}(x) + \mathcal{O}(a^6). \tag{2.23}$$

Inserting this into Eq. (2.20) and using $a^4 \sum_x \to \int \mathrm{d}^4 x$ gives the naive continuum limit

$$S_W[U] \longrightarrow \frac{1}{2g_0^2} \int \mathrm{d}^4 x\ \mathrm{tr}\, F_{\mu\nu}(x) F_{\mu\nu}(x). \tag{2.24}$$

This is the Euclidean Yang–Mills action with the bare coupling g_0.

At finite lattice spacing the Wilson action differs from the continuum action by cutoff effects. For the pure Wilson gauge action, the leading gauge-invariant discretization effects in many on-shell quantities are of order a^2. In numerical data, however, the the continuum limit is also be affected by statistics, finite-volume effects, the definition of the measured observable, and the range of available lattice spacings.

Equation (2.21) is the gauge-invariant reference theory for this study. The gauge-breaking deformation introduced in Chapter 3 is added to this action, with $\varepsilon_1 = 0$ corresponding to the undeformed Wilson theory.

2.2.4 Path Integral and Expectation Values

The lattice path integral is finite-dimensional at finite lattice spacing and finite volume. The partition function is

$$Z = \int \prod_{x,\mu} \mathrm{d}U_\mu(x)\, e^{-S[U]}, \tag{2.25}$$

where $\mathrm{d}U_\mu(x)$ is the Haar measure on SU(2). The expectation value of an observable $\mathcal{O}[U]$ is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \prod_{x,\mu} \mathrm{d}U_\mu(x)\, \mathcal{O}[U] e^{-S[U]}. \tag{2.26}$$

Gauge-invariant expectation values can be computed without gauge fixing. This is one of the practical advantages of the compact Wilson formulation.

In Monte Carlo simulations the path integral is estimated by generating an ensemble of gauge-field configurations distributed according to $e^{-S[U]}$. For an ensemble of N_{cfg} configurations U_i, the estimator is

$$\langle \mathcal{O} \rangle \approx \frac{1}{N_{\mathrm{cfg}}} \sum_{i=1}^{N_{\mathrm{cfg}}} \mathcal{O}[U_i]. \tag{2.27}$$

Because successive configurations in a Markov chain are correlated, autocorrelations must be considered when estimating uncertainties.

2.2.5 Continuum Limit and Renormalization

The continuum limit is not obtained by simply decreasing a at fixed bare coupling. Instead, the bare parameters have to be tuned so that physical observables remain fixed in physical units. For pure SU(2) gauge theory with the Wilson action, decreasing the lattice spacing corresponds to increasing β, or equivalently decreasing the bare coupling g_0.

A useful way to formulate the continuum limit is in terms of a physical length scale, such as the Sommer scale r_0. The dimensionless ratio $\frac{r_0}{a}$ is measured on the lattice. Increasing $\frac{r_0}{a}$ means that the same physical length is resolved by more lattice points, i.e. the lattice spacing is smaller in physical units. Therefore continuum extrapolations can be expressed as fits in $\left(\frac{a}{r_0}\right)^2$.

For observables derived from Wilson loops, this means that a loop size R on a coarse ensemble and the same integer R on a fine ensemble do not represent the same physical distance. Therefore, comparisons between ensembles should be made at fixed ratios such as r/r_0, or at another fixed dimensionless distance. This fixed-distance requirement will be used later when Wilson-loop observables are compared across different lattice spacings.

2.3 Static Potential and Confinement

2.3.1 Wilson Loops

The plaquette introduced above is the smallest closed product of link variables. Wilson loops generalize this construction to larger closed contours. In this thesis they are used as the basic measured observables from which the static potential and Creutz ratios are constructed.

For a straight path of length L in direction μ, define the ordered transporter

$$U_\mu(x;L) = U_\mu(x)U_\mu(x+a\hat{\mu})\cdots U_\mu(x+(L-1)a\hat{\mu}). \tag{2.28}$$

Here L is an integer lattice extent. The corresponding physical length is La.

A rectangular Wilson loop in the $\mu\nu$-plane with lattice side lengths L_μ and L_ν is defined as

$$W_{\mu\nu}(L_\mu, L_\nu) = \frac{1}{N_c V}\sum_x \operatorname{Re}\operatorname{tr}\left[U_\mu(x;L_\mu)\, U_\nu(x+L_\mu a\hat{\mu};L_\nu)\, U_\mu^\dagger(x+L_\nu a\hat{\nu};L_\mu)\, U_\nu^\dagger(x;L_\nu)\right]. \tag{2.29}$$

The factor $1/N_c$ normalizes the trace, and the sum over x averages over all lattice base points. The trace makes the closed loop gauge invariant, as explained in the discussion of closed link products above.

Using the convention that direction 4 denotes Euclidean time, the temporal Wilson loops relevant for the static potential are the loops in the $i4$-plane. For spatial direction $i = 1, 2, 3$, spatial extent R, and temporal extent T, define

$$W_{i4}(R,T) = \frac{1}{N_c V}\sum_x \operatorname{Re}\operatorname{tr}\left[U_i(x;R)\, U_4(x+Ra\hat{\imath};T)\, U_i^\dagger(x+Ta\hat{4};R)\, U_4^\dagger(x;T)\right]. \tag{2.30}$$

The Wilson loop used in the numerical analysis is the average over the three spatial orientations,

$$W(R,T) = \frac{1}{3}\sum_{i=1}^{3} W_{i4}(R,T). \tag{2.31}$$

For the SU(2) simulations considered here, $N_c = 2$, so the overall normalization is $1/(6V)$. In the following, $W(R,T)$ denotes this normalized, site-averaged and orientation-averaged temporal Wilson loop. The corresponding physical spatial and temporal sizes are

$$r = aR, \qquad t = aT. \tag{2.32}$$

Temporal Wilson loops can be interpreted as Euclidean correlation functions of static colour sources separated by the distance $r = aR$. Their spectral decomposition has the form

$$W(R,T) = \sum_n c_n(R) \exp[-aE_n(R)T], \tag{2.33}$$

where the coefficients $c_n(R)$ describe the overlap with states in the static quark–antiquark sector. At sufficiently large T, excited-state contributions are exponentially suppressed and the leading term is governed by the lowest energy,

$$W(R,T) \propto \exp[-aV(R)T]. \tag{2.34}$$

Here $V(R)$ denotes the static potential, up to the usual additive self-energy constant.

In practice, large Wilson loops are statistically noisy because the signal decreases rapidly with the loop area. Smoothing procedures such as Wilson flow can improve the statistical behaviour of measured loops. The Wilson loops defined in this subsection are the input for the static-potential analysis and for the Creutz ratios introduced below.

2.3.2 Wilson Flow and Flow-Time Convention

Wilson flow is a deterministic smoothing transformation applied to a gauge configuration after it has been generated. It introduces an auxiliary flow time t_f and a flowed link field $V_\mu(x, t_\mathrm{f})$, with initial condition $V_\mu(x, 0) = U_\mu(x)$. Note that the flow time t_f is not the temporal extent T of a Wilson loop. The variable T denotes the size of an observable in the Euclidean time direction, whereas t_f controls the amount of smoothing applied to the gauge field before the observable is measured.

Wilson flow is the lattice gradient flow generated by the Wilson plaquette action. It evolves the link variables according to

$$\partial_{t_\mathrm{f}} V_\mu(x, t_\mathrm{f}) = Z_\mu(x, t_\mathrm{f})\, V_\mu(x, t_\mathrm{f}), \tag{2.35}$$

where

$$Z_\mu(x, t_\mathrm{f}) = -g_0^2\, \partial_{x,\mu} S_W[V(t_\mathrm{f})]. \tag{2.36}$$

Here $\partial_{x,\mu} S_W$ denotes the derivative of the Wilson plaquette action with respect to the link $V_\mu(x, t_\mathrm{f})$, where the derivative is taken along infinitesimal SU(2) group directions. Equivalently, the resulting force $Z_\mu(x, t_\mathrm{f})$ is projected onto the Lie algebra SU(2). This ensures that the flow changes the links within the gauge group.

Equivalently, for an infinitesimal flow step one may write

$$V_\mu(x, t_\mathrm{f} + \delta t_\mathrm{f}) = \exp[\delta t_\mathrm{f} Z_\mu(x, t_\mathrm{f})]\, V_\mu(x, t_\mathrm{f}) + \mathcal{O}(\delta t_\mathrm{f}^2). \tag{2.37}$$

This form makes clear why the flowed variables remain group-valued links: each step multiplies the old link by the exponential of a Lie-algebra element.

The flow is not a Monte Carlo update and does not generate a new ensemble. Instead, each gauge configuration U is deterministically mapped to a smoothed configuration $V(t_\mathrm{f})$ for each chosen flow time t_f. Wilson-loop observables at positive flow time are obtained by replacing the original links in the definitions of Sec. 2.3.1 by flowed links. We denote the resulting observable by

$$W(R,T;t_\mathrm{f}) = W_{V(t_\mathrm{f})}(R,T), \tag{2.38}$$

so that $t_\mathrm{f} = 0$ gives the unflowed Wilson loop.

The smoothing effect of gradient flow is characterized by the smoothing radius r_sm. With the flow time specified in lattice units, $t_\mathrm{lat} = t_\mathrm{f}/a^2$, the relevant relations are

$$r_\mathrm{sm} = \sqrt{8t_\mathrm{f}}, \qquad \frac{r_\mathrm{sm}}{a} = \sqrt{8t_\mathrm{lat}}, \qquad r_\mathrm{sm} = a\sqrt{8t_\mathrm{lat}}. \tag{2.39}$$

Keeping t_lat fixed therefore means keeping the smoothing radius fixed in lattice units. Therefor in physical units the smoothing radius vanishes as $a \to 0$.

This convention is important for the continuum interpretation. At fixed t_f/a^2, Wilson flow acts as a smearing procedure: it suppresses ultraviolet fluctuations at finite lattice spacing, but its physical smearing radius shrinks in the continuum limit. The zero-flow continuum target is therefore unchanged. By contrast, keeping t_f fixed in physical units would define a genuinely flowed observable with a nonzero physical smoothing radius and hence a different continuum observable. This distinction between gradient-flow smearing and physical gradient flow is the convention used in the continuum analysis below.

2.3.3 Effective Potential Extraction

Temporal Wilson loops are used to extract the static quark–antiquark potential. For large Euclidean time extent T, the Wilson loop has the form

$$W(R,T) = \sum_n c_n(R) e^{-aE_n(R)T}, \tag{2.40}$$

where $E_0(R) = V(R)$ is the static ground-state potential and the higher $E_n(R)$ describe excited states with the same static sources. At sufficiently large T, the ground state dominates:

$$W(R,T) \propto e^{-aV(R)T}. \tag{2.41}$$

A lattice effective potential is therefore defined by the ratio

$$aV_\mathrm{eff}(R,T) = -\ln\left[\frac{W(R,T+1)}{W(R,T)}\right]. \tag{2.42}$$

If excited-state contributions are negligible, this approaches the desired static potential in lattice units:

$$aV_\mathrm{eff}(R,T) \xrightarrow[T\to\infty]{} aV(R). \tag{2.43}$$

2.3.4 Creutz Ratios

Creutz ratios are combinations of neighboring Wilson loops designed to cancel some perimeter and normalization contributions. In the continuum they can be viewed as the mixed derivative

$$\chi(r,t) = -\frac{\partial^2}{\partial r\,\partial t}\ln W(r,t), \tag{2.44}$$

where $W(r,t)$ is a rectangular Wilson loop of physical spatial extent r and Euclidean time extent t. A central finite-difference discretization gives a lattice Creutz ratio located at half-integer separations,

$$\chi\left(R+\frac{1}{2}, T+\frac{1}{2}\right) = \ln\left[\frac{W(R+1,T)W(R,T+1)}{W(R,T)W(R+1,T+1)}\right], \tag{2.45}$$

The normalization of the Wilson loop, including the factor $1/N_c$, cancels in this ratio. This is useful for the present SU(2) analysis because the same formula can be applied directly to the normalized Wilson-loop output described above.

The central-difference form approximates the continuum mixed derivative with cutoff effects of order a^2 when the underlying Wilson loop is smooth. This makes Creutz ratios particularly useful for continuum studies: they are finite observables, their leading normalization and perimeter contributions cancel, and they are closely related to the static force. In the large-time limit,

$$\frac{\chi(R,T)}{a^2} \xrightarrow[T\to\infty]{} F(r) \tag{2.46}$$

up to the usual distinction between the discrete lattice midpoint and the physical distance r. Diagonal ratios $\chi(R,R)$ are also useful summary observables because they probe the same spatial and temporal scale and are straightforward to compare across ensembles [4, 5].

If the large Wilson loops are dominated by an area law, their leading behaviour may be written as

$$W(R,T) \sim \exp\Big[-\sigma a^2 RT - \mu a(R+T) - c\Big], \tag{2.47}$$

where R and T are lattice extents, so that the physical area is $(Ra)(Ta) = a^2RT$. The coefficient μ denotes a perimeter contribution and c is a constant.

Inserting Eq. (2.47) into the Creutz-ratio definition shows that the perimeter and constant terms cancel [maybe also source]. The remaining area term gives

$$\chi(R,T) \xrightarrow[R,T\to\infty]{} \sigma a^2. \tag{2.48}$$

Thus Creutz ratios provide a direct estimator of the string tension in lattice units, $a^2\sigma$, up to finite-size, excited-state, and lattice-spacing corrections.

At finite R and T, Creutz ratios contain corrections from excited states, Coulombic contributions, finite-volume effects, and lattice artifacts. They are therefore best interpreted either as approximate string-tension estimators at large loop sizes or as diagnostic observables for scaling comparisons between ensembles.

For continuum extrapolations the important distinction is between fixed lattice separation and fixed physical distance. Holding R fixed while taking $a \to 0$ drives the physical distance $r = Ra$ to zero and therefore probes the short-distance limit rather than the same physical force. The continuum comparison therefore uses a dimensionless distance

$$\hat{r} = \frac{r}{r_0} \tag{2.49}$$

and interpolates the measured Creutz ratios to common $\hat{r}$ values before fitting in a/r_0. This is also where gauge-breaking effects are expected to be most visible: short-distance Wilson-loop observables are sensitive to ultraviolet deformations, while large loops suffer from rapidly increasing statistical noise.

2.3.5 Sommer Scale and String Tension

The static potential can also be used to set the scale. [A common phenomenological form[rephrase]] is the Cornell-type potential

$$V(r) = V_0 - \frac{\alpha}{r} + \sigma r, \tag{2.50}$$

where V_0 is an additive self-energy, α parametrizes the short-distance Coulombic part, and σ is the string tension. The corresponding force is

$$F(r) = \frac{\mathrm{d}V}{\mathrm{d}r}. \tag{2.51}$$

The additive constant V_0 drops out of the force.

The Sommer scale r_0 is defined implicitly by

$$r_0^2 F(r_0) = 1.65. \tag{2.52}$$

This definition uses the force at intermediate distances, where the potential can often be determined more accurately than at very large distances. In lattice units, the force is estimated from finite differences of the potential. If $\hat{V}(R) = aV(Ra)$, then

$$\hat{F}\left(R + \frac{1}{2}\right) = \hat{V}(R+1) - \hat{V}(R) \tag{2.53}$$

is a dimensionless estimate of $a^2 F(r)$ at an intermediate distance. The Sommer condition can then be written as

$$\left(\frac{r}{a}\right)^2 \hat{F}\left(\frac{r}{a}\right) = 1.65. \tag{2.54}$$

Solving this equation gives r_0/a.

The ratio r_0/a is one of the main scale-setting quantities in this thesis. It allows data at different β values to be compared in approximately physical units and provides the continuum variable

$$\frac{a}{r_0} = \left(\frac{r_0}{a}\right)^{-1}. \tag{2.55}$$

The continuum limit corresponds to

$$\frac{a}{r_0} \to 0. \tag{2.56}$$

The string tension may also be expressed in dimensionless form as

$$\sigma r_0^2. \tag{2.57}$$

This quantity is useful because it is independent of the arbitrary choice of lattice units. Agreement of such dimensionless quantities along a sequence of lattice spacings is a central test of scaling toward a common continuum limit.

Summary and Connection to the Deformation

[todo improve] This chapter introduced the theoretical and numerical ingredients used in the rest of this thesis. The Wilson plaquette action is the gauge-invariant part of the lattice action and defines the undeformed theory at $\varepsilon_1 = 0$. Wilson loops, Creutz ratios, the static potential, and the Sommer scale define the observables used to compare ensembles at different lattice spacings. Wilson flow is used as a smoothing procedure at fixed flow time in lattice units, so that its physical smoothing radius vanishes in the continuum limit.

The next chapter adds a controlled single-link term to the Wilson action. Unlike the plaquette term, this additional term is not invariant under local gauge transformations. The purpose of the following analysis is therefore to test whether the effects of this explicit gauge breaking disappear, persist, or modify the continuum extrapolation of gauge-invariant observables.

3 Gauge-Breaking Deformation

This chapter defines the explicit gauge-breaking deformation used in the simulations and gives the scaling estimate that motivates the numerical tests. The purpose is not to construct a new continuum theory, but to test whether a controlled violation of gauge invariance can be suppressed as the lattice spacing is reduced.

3.1 Definition of the Deformation

The gauge-invariant reference theory is pure SU(2) lattice gauge theory with the Wilson plaquette action. The deformed theory is obtained by adding a linear single-link term. Up to additive constants that do not affect normalized expectation values, the action used in the simulations is

$$S[U] = -\beta \sum_{x} \sum_{\mu<\nu} \frac{1}{N_c} \operatorname{Re}\operatorname{tr} U_{\mu\nu}(x) - \varepsilon_1 \sum_{x} \sum_{\mu=1}^{4} \frac{1}{N_c} \operatorname{Re}\operatorname{tr} U_{\mu}(x). \tag{3.1}$$

Here $U_{\mu\nu}(x)$ is the plaquette matrix, $N_c = 2$ for the ensembles considered in this thesis, and ε_1 controls the strength of the explicit gauge-breaking deformation. The undeformed Wilson theory is recovered for

$$\varepsilon_1 = 0. \tag{3.2}$$

For positive ε_1, the single-link contribution lowers the action when $\operatorname{Re}\operatorname{tr} U_{\mu}(x)$ is large. The deformation therefore biases individual links toward the identity element. This provides a simple controlled way to violate local gauge invariance while keeping the deformation strength tunable.

3.2 Breaking of Local Gauge Invariance

As discussed in Sec. 2.2.2, closed products of link variables transform by conjugation under local gauge transformations. Therefore the plaquette trace $\operatorname{Re}\operatorname{tr} U_{\mu\nu}(x)$ is gauge invariant, and the Wilson part of Eq. (3.1) preserves local gauge invariance.

The new term in Eq. (3.1) is different because it contains a single open link. Under a local lattice gauge transformation,

$$U_{\mu}(x) \mapsto \Omega(x) U_{\mu}(x) \Omega^{\dagger}(x + a\hat{\mu}), \tag{3.3}$$

and therefore the single-link trace transforms as

$$\operatorname{Re}\operatorname{tr} U_{\mu}(x) \mapsto \operatorname{Re}\operatorname{tr} \left[\Omega(x) U_{\mu}(x) \Omega^{\dagger}(x + a\hat{\mu})\right]. \tag{3.4}$$

Because the two group elements at the endpoints of the link are independent, the transformed trace is not equal to the original trace in general. Thus any nonzero ε_1 explicitly breaks local gauge invariance.

A global diagonal SU(2) symmetry remains. If $\Omega(x) = \Omega$ is the same group element at every site, then

$$U_\mu(x) \mapsto \Omega U_\mu(x) \Omega^\dagger, \tag{3.5}$$

and the trace is unchanged. The deformation therefore breaks local gauge invariance while preserving global SU(2) conjugation symmetry.

This deformation should not be interpreted as a gauge fixing of the Wilson theory. Gauge fixing selects representatives of gauge orbits, whereas the single-link term changes the Boltzmann weight itself. Configurations related by a local gauge transformation no longer have the same statistical weight.

3.3 Weak-Field Scaling Estimate

To estimate the leading scaling behaviour of the deformation, it is sufficient to expand the single-link term for smooth fields close to the continuum limit. Using the weak-field relation between a link variable and the continuum gauge field introduced in Sec. 2.2.2,

$$U_\mu(x) = \exp[iag_0 A_\mu(x)]\,, \qquad A_\mu(x) = A_\mu^a(x) T^a. \tag{3.6}$$

Expanding the single-link trace around the identity for SU(2) gives,

$$\frac{1}{2}\operatorname{Re}\operatorname{tr} U_\mu(x) = 1 - \frac{a^2 g_0^2}{4}\operatorname{tr} A_\mu^2(x) + \mathcal{O}(a^4 A_\mu^4). \tag{3.7}$$

Only the leading scaling is needed in the following: the first nonconstant contribution is of order $a^2 g_0^2 A_\mu^2$.

Inserting this expansion into the single-link part of Eq. (3.1) gives

$$S_{\text{break}} = \text{const.} + \frac{\varepsilon_1 g_0^2}{4a^2}\int \mathrm{d}^4x \sum_\mu \operatorname{tr} A_\mu^2(x) + \cdots\,. \tag{3.8}$$

Using $\beta = \frac{4}{g_0^2}$ for SU(2), the coefficient of this leading contribution is proportional to

$$\frac{\varepsilon_1}{\beta a^2}. \tag{3.9}$$

The operator $\sum_\mu \operatorname{tr} A_\mu^2$ is not gauge invariant. It is forbidden in the undeformed Wilson theory by local gauge symmetry, but the single-link deformation removes this symmetry protection. Since A_μ has canonical dimension one in four dimensions, the leading contribution has the form of a dimension-two gauge-noninvariant masslike term. It is therefore relevant by power counting.

This estimate explains why a fixed nonzero value of ε_1 should not be treated as an ordinary irrelevant cutoff effect. A dimensionless measure of the induced coefficient in units of the Sommer scale is

$$\bar{\varepsilon} = \frac{\varepsilon_1}{\beta}\left(\frac{r_0}{a}\right)^2. \tag{3.10}$$

At fixed ε_1, this quantity grows as $a \to 0$, up to the logarithmic running contained in β. Thus a fixed bare breaking parameter does not correspond to a fixed physical breaking strength.

3.4 Scaling of the Breaking Parameter

To approach the gauge-invariant continuum theory, the dimensionless breaking strength in Eq. (3.10) should be suppressed as the lattice spacing is reduced. This motivates a scaling prescription for ε_1.

Let $\beta^\star$ denote a reference ensemble with breaking parameter $\varepsilon_1^\star$ and measured scale $(r_0/a)^\star$. The scaled trajectory is defined by

$$\varepsilon_1(\beta) = \varepsilon_1^\star \frac{\beta}{\beta^\star}\left[\frac{(r_0/a)^\star}{r_0/a(\beta)}\right]^{2+p}. \tag{3.11}$$

The power 2 compensates the engineering dimension of the masslike coefficient in Eq. (3.9). The additional exponent p controls whether the induced breaking strength is kept approximately fixed or driven to zero in physical units.

Combining Eq. (3.11) with Eq. (3.10) gives

$$\bar{\varepsilon}(\beta) = \bar{\varepsilon}^\star\left[\frac{(r_0/a)^\star}{r_0/a(\beta)}\right]^{p}. \tag{3.12}$$

Thus $p = 0$ keeps the estimated induced coefficient approximately fixed in physical units, while $p > 0$ suppresses it as the lattice spacing is reduced. The main scaled trajectory used in this thesis is $p = 1$. On this trajectory, $\bar{\varepsilon}$ decreases linearly in a/r_0.

The fixed-ε_1 ensembles therefore serve as stress tests of a relevant gauge-breaking deformation. The scaled ensembles test whether the Wilson continuum limit is recovered when the relevant gauge-breaking contribution is suppressed.

3.5 Implications for the Continuum Analysis

The observables measured in this thesis are gauge invariant, but their expectation values are computed with the deformed probability weight $e^{-S[U]}$. They can therefore depend on ε_1 at nonzero lattice spacing. Such dependence is expected and is not by itself evidence for a different continuum limit.

The relevant diagnostic is the continuum intercept. Let $\hat{O}$ denote a dimensionless observable, for example σr_0^2 or a Creutz ratio evaluated at fixed physical distance.

Restoration of the Wilson continuum limit means that the undeformed Wilson ensembles and the scaled gauge-broken ensembles approach the same continuum value,

$$\hat{O}(a, \varepsilon_1) \xrightarrow[a\to 0]{} O_0. \tag{3.13}$$

In this case the gauge-breaking term changes only the cutoff dependence in the range of lattice spacings studied.

If, instead, the deformed ensembles approach a different continuum intercept, then the deformation has not been scaled away in the fitted range. The continuum analysis therefore distinguishes

$$\text{common continuum intercept} \quad \Longleftrightarrow \quad \text{restoration of the Wilson continuum limit}, \tag{3.14}$$

from

$$\text{shifted continuum intercept} \quad \Longleftrightarrow \quad \text{persistent effect of the gauge-breaking deformation}. \tag{3.15}$$

This is the central diagnostic used in the following chapters. Fixed ε_1 tests the sensitivity of the theory to a relevant gauge-breaking deformation, while scaled ε_1 tests whether this deformation can be suppressed sufficiently to recover the gauge-invariant continuum theory. [check if both is used]

4 Methodological Approach

This chapter describes the numerical procedure used to test whether the gauge-breaking deformation changes the continuum limit. The analysis consists of four steps: generation of Wilson and deformed gauge ensembles, measurement of Wilson-loop observables, statistical error estimation, and continuum comparison at fixed physical distance.

4.1 Ensemble Organization and Fixed-Physics Comparisons

Ensembles are grouped by their bare parameters (β, ε_1). Multiple runs with identical parameters are treated as statistically independent streams after thermalization and are combined at the level of bootstrap samples. Since the lattice spacing changes with β, observables are not compared at fixed integer loop size R, but at fixed dimensionless distance r/r_0.

4.2 Parameter Trajectories

The simulations use two types of trajectories in the bare parameter space. The undeformed Wilson ensembles with $\varepsilon_1 = 0$ define the reference theory. Fixed-ε_1 ensembles test the sensitivity to an unsuppressed gauge-breaking deformation. Scaled-ε_1 ensembles implement the scaling prescription of Chapter 3 and test whether the Wilson continuum limit is recovered when the relevant contribution is suppressed.

4.2.1 Thermalization Cuts

Thermalization cuts are chosen separately for each ensemble, because the equilibration time depends on the bare parameters β and ε_1. In particular, changing the lattice spacing or the strength of the gauge-breaking deformation can change how quickly the Markov chain reaches its stationary distribution.

The cuts are determined by inspecting Monte Carlo histories of the measured Wilson loops. Since Wilson loops are the primary observables used later for the static potential, Creutz ratios, and scale setting, they provide the most direct check that the relevant part of the measurement history has equilibrated. Simple observables such as the plaquette and the average link trace are used as additional cross-checks.

For each run, the initial part of the Markov chain is discarded until the Wilson-loop histories fluctuate around a stationary mean without visible drift. Runs with the same

β	$N_s^3 \times N_4$	ε_1	trajectory	sweeps	measurements

Table 4.1: Overview of the ensembles used in this thesis.

(β, ε_1) are combined only after applying the corresponding thermalization cut to each individual stream.

4.2.2 Bootstrap Blocking

The measurements after thermalization are still not statistically independent, because consecutive configurations are generated by a Markov chain. A naive bootstrap over individual measurements would therefore underestimate the statistical error. To account for this, the ordered measurement history is divided into contiguous blocks of fixed length b in simulation time. The bootstrap samples are then generated by drawing these blocks with replacement and averaging over the resampled history. Unless stated otherwise, the statistical errors quoted in this work use $N_{\text{boot}} = 1000$ bootstrap replicas.

The block length must be large compared with the autocorrelation time of the observables used in the analysis. If b is too small, correlations remain between the resampled data and the error estimate is biased downward. If b is too large, the number of available blocks becomes small and the bootstrap estimate becomes noisy. In practice, several block sizes are compared and the chosen value is taken from the region where the estimated errors are stable under further increases of b. [add an example plot of that]

The same bootstrap samples are propagated through the nonlinear analysis steps. Wilson loops are resampled first, and derived quantities such as effective potentials, the static potential $V(R)$, Creutz ratios, and the Sommer scale are recomputed for each bootstrap replica. This preserves correlations between quantities derived from the same ensemble and gives the statistical uncertainties used in the later continuum analysis.

4.2.3 Integrated Autocorrelation Time

The block size is chosen such that the estimated statistical error becomes stable when the block size is increased. As a cross-check, the integrated autocorrelation time of simple observables is used to verify that the block length is larger than the dominant autocorrelation scale.

4.2.4 ensemble table

4.3 Analysis Pipeline and Data Reduction

The analysis starts from individual simulation runs. Runs with identical base parameters, in particular the same β, lattice size, and ε_1, are assigned to the same ensemble. They may differ in random seed or total statistics, but are treated as independent Markov-chain streams of the same lattice theory.

For each run, Wilson-loop histories are inspected and an ensemble-specific thermalization cut is chosen. Measurements before this cut are discarded. The remaining measurements from runs with identical base parameters are combined only after applying the corresponding cut to each stream.

The combined post-thermalization data form the input for the later analysis steps. Wilson loops are used to extract effective potentials, Creutz ratios, and the Sommer scale r_0/a. The measured scale is then used to express observables at fixed dimensionless distance r/r_0 and to form the continuum variable a/r_0. For deformed ensembles, the same scale is also used to construct the dimensionless breaking estimate

$$\bar{\varepsilon} = \frac{\varepsilon_1}{\beta}\left(\frac{r_0}{a}\right)^2 .$$

The final output of the pipeline is therefore one summarized data set per ensemble, containing the Wilson-loop observables, derived quantities, statistical uncertainties, and the dimensionless variables required for the continuum comparison.

4.4 Extraction of Physical Observables

4.4.1 Wilson-Loop Analysis

effective-mass plateaus, the choosing the right fit windows. Results in systematic uncertainty because of varying those windows.

4.4.2 Static Potential

Choosing an r_{min} for the cornel fit. Looking at χ.

4.4.3 Creutz-Ratio Analysis

The measured Wilson loops are converted into effective potentials and Creutz ratios using the definitions of Secs. ... The potential analysis is used for scale setting through r_0/a, while Creutz ratios are used for direct continuum comparisons at fixed r/r_0.

4.4.4 Scale Setting

Scale setting at $\varepsilon_1 = 0$

4.5 Continuum Extrapolation Strategy

For each observable $\hat{O}$, values are compared at fixed physical distance or fixed dimensionless scale. The continuum extrapolation is performed as a function of $(a/r_0)^2$, using the undeformed Wilson ensembles as the reference. The main test is whether the Wilson and scaled deformed ensembles are compatible with a common continuum intercept.

4.6 Continuum Extrapolation and Scaling Tests

4.6.1 Choice of Continuum Variable

The primary continuum variable in this thesis is the measured dimensionless lattice spacing

$$\hat{a} = \frac{a}{r_0}. \tag{4.1}$$

For observables that are expected to have leading even-power cutoff effects with the Wilson gauge action, the default horizontal axis is therefore $\hat{a}^2 = (a/r_0)^2$. Using the measured scale rather than the bare coupling makes the extrapolation closer to a fixed-physics comparison and reduces the risk of interpreting changes in scale setting as physical changes in the observable itself.

For Creutz-ratio observables the quantity to extrapolate should be dimensionless. Convenient choices are

$$\hat{\chi}(\hat{r}) = r_0^2 \frac{\chi(r,t)}{a^2}, \qquad \hat{r} = \frac{r}{r_0}, \tag{4.2}$$

or, for large loops where the string-tension interpretation is appropriate, σr_0^2. The measured lattice ratio $\chi(R,T)$ is dimensionless and approaches σa^2 in the area-law regime. Deviding by a^2 and multipying by r_0^2 converts it to a quantity with a finite physical continuum limit. The same logic applies to the force extracted from the static potential.

4.6.2 Fixed-Physics Trajectories

Ensembles at different β values are compared along trajectories where the physical volume and the dimensionless analysis point are kept as fixed as the available data allow. In practice this means comparing observables at common $\hat{r} = r/r_0$ and using only ensembles whose volumes are large enough that finite-volume corrections are not expected to dominate the Wilson-loop signal. When exact matching is not possible, the mismatch is included as a systematic effect.

The gauge-breaking parameters are treated as additional bare deformation parameters. If they are held fixed while a changes, the continuum fit is a stress test of the relevant deformation identified in Eq. (3.8). Fixed bare ε_1 makes $\bar{\varepsilon}$ grow approximately as $(r_0/a)^2/\beta$, so it is not the natural asymptotic restoration trajectory. When the scaled prescription (3.11) is used, the fit must include the chosen value of p. The case $p = 0$ keeps the induced masslike coefficient approximately fixed in physical units, while the main choice $p = 1$ drives it to zero linearly in $\hat{a}$. [repetition]

4.7 Computational Environment

The simulations were carried out on the QBig cluster. Most production runs were performed on nodes 13–17 equipped with NVIDIA P100 GPUs. Additional and faster runs were performed on nodes 18 and 19 equipped with NVIDIA A100 GPUs. The GPU type affects the wall-clock runtime but not the generated Markov-chain ensemble.

5 Results

5.1 Thermalization

5.2 Gauge-Invariant Baseline

$\varepsilon = 0$ results for $W(R, T)$, $V(R)$, r_0, and σ

5.3 Continuum Limit at Fixed Breaking

6 Discussion

6.1 Interpretation of the Continuum Limit

6.2 Comparison with Theoretical Expectations

Compare to Chapter 3.

6.3 Limitations

6.4 Implications

7 Conclusion and Outlook

7.1 Summary

7.2 Answer to the Research Question

7.3 Future Work

How could this be improved.

A Simulation Tables

B Additional Plots

C Analysis Code and Reproducibility

[Code in github, maybe a short code description]

D Derivations

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